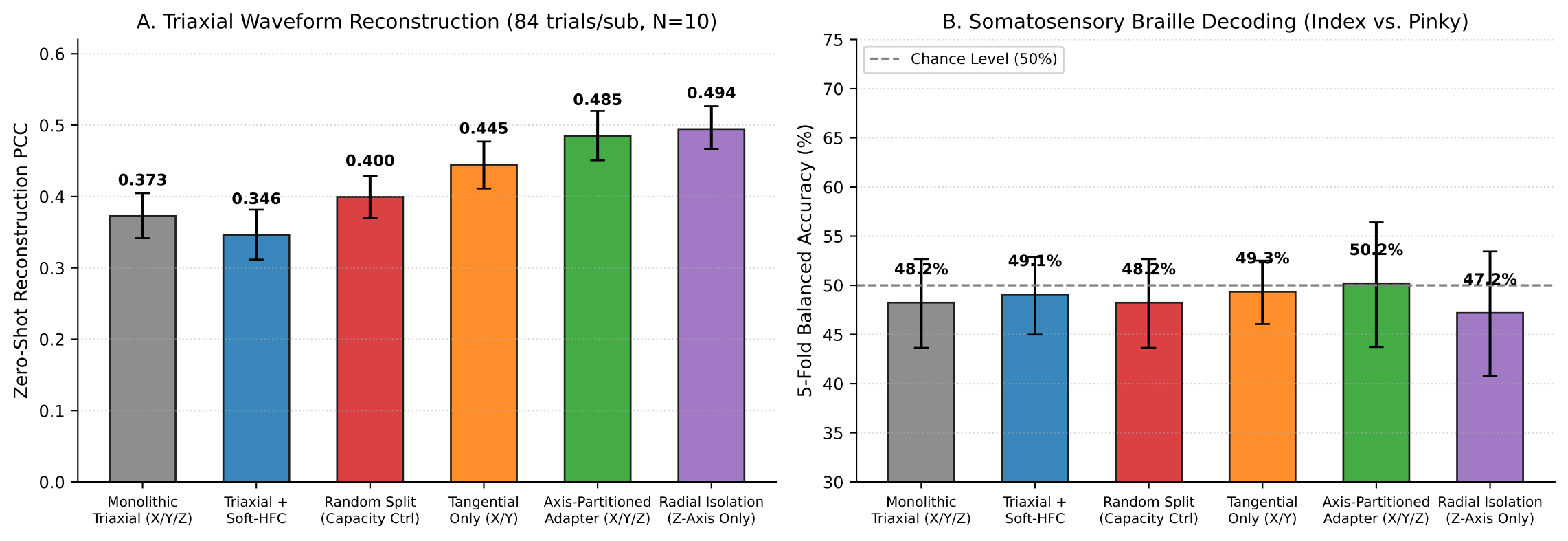
A. Triaxial Waveform Reconstruction (84 trials/sub, N=10)
Zero-Shot Reconstruction PCC
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.373
0.346
0.400
0.445
0.485
0.494
Monolithic Triaxial (X/Y/Z)
Triaxial + Soft-HFC
Random Split (Capacity Ctrl)
Tangential Only (X/Y)
Axis-Partitioned Adapter (X/Y/Z)
Radial Isolation (Z-Axis Only)
B. Somatosensory Braille Decoding (Index vs. Pinky)
5-Fold Balanced Accuracy (%)
Chance Level (50%)
30
35
40
45
50
55
60
65
70
75
48.2%
49.1%
48.2%
49.3%
50.2%
47.2%
Monolithic Triaxial (X/Y/Z)
Triaxial + Soft-HFC
Random Split (Capacity Ctrl)
Tangential Only (X/Y)
Axis-Partitioned Adapter (X/Y/Z)
Radial Isolation (Z-Axis Only)